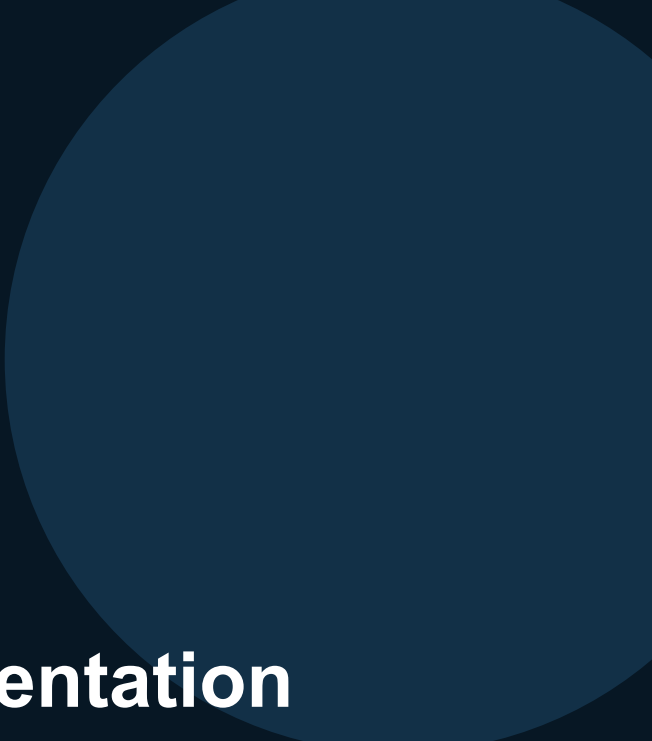




QUANT'S TRADE

Current Implementation Feature Inventory

Only capabilities evidenced in the application repository.
No planned or aspirational features included.

REPOSITORY-AUDITED EDITION / 21 JULY 2026



Scope of This Document

This inventory describes functionality represented by routed UI pages, API endpoints, database models, services, Python modules, middleware and automated tests in the current repository. It does not claim that optional providers are configured, that every feature is deployed in production, or that live broker execution is certified.

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1. Platform Foundation

Authentication

- Email and password registration
- Password hashing
- Login and logout
- Short-lived access tokens
- Refresh-token persistence and rotation
- HttpOnly authentication cookies
- CSRF token endpoint and mutation protection
- Authenticated current-user endpoint
- Per-route authentication middleware
- Login and registration rate limiting

Users and Access

- Three user roles: user, admin and owner-admin
- Pending, verified, rejected and suspended account states
- Verified-user enforcement
- Page-level access permissions
- User ownership validation
- Admin-mode toggle
- Read-only demo access
- Demo-session reset
- Per-user application settings

Persistence

- PostgreSQL data model
- Prisma ORM
- Versioned database migrations
- Repository and service data-access layers
- User-scoped records
- Transactions for ledger and portfolio operations
- Persisted audit records
- JSON fallback support for selected runtime data
- Database seed and backfill scripts

Caching and Runtime

- Upstash Redis market-data cache provider
- In-memory cache fallback
- TTL-based cache entries
- Market-data range and interval freshness policies
- Frontend request deduplication
- Coordinated polling
- User-scoped Python runtime directories
- Persisted engine and scan runtime artifacts

2. Application Experience

Web Application

- React 19 frontend
- Vite build system
- Lazy-loaded application pages
- Feature-based frontend organization
- Fetch-based API client
- Shared request manager
- Loading, error and empty states
- Confirmation dialogs
- Reusable cards, metrics, charts and progress components

Navigation and Layout

- Trading cockpit application shell
- Collapsible desktop sidebar
- Responsive mobile drawer
- Mobile bottom navigation
- Page-aware navigation
- Role-aware visible pages

- User identity panel
- Admin-mode indicator
- Dark and light theme switching
- Persisted sidebar and theme preferences

Current Pages

- Landing page
- Login and registration
- Interactive demo
- Dashboard
- Scanner
- Watchlist
- Strategy Lab
- Approvals
- Portfolio
- Research
- Validation
- Playbook
- Alerts
- Settings
- Admin Dashboard
- Broker Center
- Responsive audit page

Visual Layer

- Design-token styles for color, spacing, typography, radius and elevation
- Premium theme styles
- Desktop, tablet and mobile scanner layouts
- Responsive primitives
- Framer Motion interactions
- Recharts analytics
- Lightweight Charts stock-price visualization
- Stock workspace modal and sidebar

3. Dashboard and Runtime Control

Trading Dashboard

- Opportunity summary
- Signal distribution chart
- Top-opportunity chart
- Scanner result cards
- Portfolio headline data
- Risk dashboard data
- Broker-derived portfolio headline support
- Engine and data-health status
- Refresh orchestration across active modules

Engine Operations

- Engine status endpoint
- Start engine action
- Stop engine action
- Run scan action
- Safety status endpoint
- Data-health endpoints
- System runtime diagnostics
- Market-session-aware scheduler
- Scheduled scan persistence
- Scheduler process locking
- Scan progress artifact parsing

Trading Session

- Trading-session summary endpoint
- Execution settings
- Market-universe settings

- Trading horizon selection
- Risk multiplier resolution
- Execution-mode normalization
- Per-user runtime isolation

4. Scanner, Watchlist and Universes

Scanner Jobs

- Manual scan start
- Scheduled scan job type
- Queued, running, completed, failed, cancelled and timeout states
- Active scan lookup
- Scan-job status lookup
- Scan cancellation
- Scan history
- Persisted scan results
- Persisted opportunities
- Progress stages and processed-symbol counts

Opportunity Analysis

- BUY, HOLD and SELL signals
- Opportunity scores
- Confidence scores and confidence breakdown
- Backtest return, win-rate and drawdown inputs
- Risk-adjusted opportunity data
- Position-aware opportunity evaluation
- Regime-aware trade gating
- Signal-change history
- Per-symbol opportunity history
- Stock-detail drawer

Scanner Interface

- Table and card views
- Market, exchange and currency filters
- Watchlist-only scanning
- Filter chips
- Saved scanner views
- Scan progress panel
- Stock price charts
- Market metadata formatting
- Responsive scanner layouts

Watchlists and Universes

- User watchlist list, add/update and remove
- Scan watchlist workflow
- Manual stock universes
- Watchlist universes
- Sector universes
- Industry universes
- S&P; 500 sample universes
- Custom-screen universes
- Universe create, update and delete
- Universe member add and remove
- US and SGX candidate-symbol support

5. Python Quantitative Engine

Indicators

- Configurable fast and slow EMA
- RSI
- ATR
- Daily returns
- Rolling volatility
- Volume moving average

- Rolling high
- MACD line
- MACD signal
- MACD histogram

Signals and Strategy Evaluation

- Technical signal generation
- Normalized confidence scoring
- Structured strategy JSON validation
- Nested rule interpretation
- Indicator-to-indicator comparisons
- Crossovers
- Between conditions
- Volume-spike conditions
- Risk-rule gating
- Strategy-specific indicator configuration

Regime and Horizon

- Row-level market-regime classification
- Trade summaries by regime
- Macro regime using SPY trend, VIX, ten-year yield and inflation proxy
- Risk-on and risk-off controls
- Regime risk multipliers
- Allowed strategy types by regime
- Trading-horizon profiles
- Horizon-specific scan intervals
- Horizon-specific indicator and signal settings

News and Events

- Fintnhub company-news provider
- News event mapping
- Earnings, macro, Fed and SEC-filing event contracts
- Provider result validation
- Provider latency and success metrics
- High-risk news checks
- Near-earnings checks
- Macro-event checks
- Optional AI news reasoning through the Node AI gateway
- Fail-safe reasoning fallback

Position Awareness

- Portfolio-value calculation
- Existing-position lookup
- Proposed order construction
- Averaging detection
- Stop-loss risk calculation
- Exposure-aware quantity adjustment
- Sector exposure checks
- Horizon compatibility
- Portfolio-fit evaluation

6. Strategy Lab

Experiments

- Strategy experiment list and creation
- Experiment update and deletion
- Strategy preview
- Persisted experiment settings
- Experiment execution
- Persisted strategy runs and trades
- Active strategy selection
- Strategy leaderboard
- Strategy memory view

Strategy DSL

- Shared strategy JSON contract
- Backend DSL validator
- Python contract validator
- Strategy compiler
- Python strategy interpreter
- Entry, exit and risk rule representation
- Nested groups and comparisons
- Compiled strategy execution routing
- Malformed and unsupported rule rejection

AI Strategy Copilot

- Generate strategy drafts
- Explain strategies
- Review strategies
- Answer strategy questions
- Generate strategy research reports
- Answer research questions
- Compare strategy versions
- Propose edits
- Approve edits before application
- Evidence-aware copilot context

Lifecycle

- Draft, research, candidate, active and archived deployment states
- Strategy versions
- Deployment-readiness calculation
- Lifecycle dashboard
- Active strategy set
- Assign and remove active strategies
- Activate and deactivate deployment versions
- Validation aggregates
- Lifecycle evidence

7. Backtesting and Strategy Evidence

Backtesting

- Single-symbol backtesting
- Multi-asset backtesting
- Shared-capital portfolio backtesting
- Date-range validation
- Initial cash and risk-per-trade controls
- Strategy-specific signal generation
- Volume participation caps
- Slippage and spread assumptions
- Commission and regulatory-fee assumptions
- Trade and equity-curve persistence

Performance Metrics

- Total return
- CAGR
- Maximum drawdown
- Sharpe ratio
- Sortino ratio
- Annualized volatility
- Profit factor
- Expectancy
- Average holding period
- Completed-trade count
- Exposure statistics
- Annual and monthly returns
- Calmar ratio in matrix replay

Robustness

- Parameter sweeps
- EMA and RSI perturbation tests
- Execution-cost stress variant
- Date-window shift test
- Parameter-sensitivity score
- Return confidence range
- Stable, moderate and fragile ratings
- Overfitting risk detection
- Train/test divergence checks
- Trade-count and drawdown stability checks

Walk Forward and Stress

- Walk-forward window generation
- Train and test segments
- Multiple walk-forward modes
- Persisted segments and metrics
- Regime analysis
- Strategy stress testing
- Monte Carlo trade-order shuffling
- Slippage and partial-fill simulation
- Risk-of-ruin estimate
- Probability of 20% drawdown
- Worst, median and best simulated return

Validation

- Persisted validation signals
- Pending, matured, failed and skipped evaluation states
- Signal evaluation action
- Confidence calibration view
- Regime validation view
- Sector validation view
- OpenAI-impact view
- Validation data refresh

8. Strategy Matrix and Allocation

Deployment Matrix

- Strategy, sector and regime deployment routes
- Strategy deployment sets
- Active, pending, sit-out and blocked allocation cells
- Equal-weight allocation
- Manual-weight allocation
- Risk-parity allocation
- Confidence-weighted allocation
- Evidence-weighted allocation
- Volatility-adjusted allocation
- Allocation snapshots and audit logs

Allocation Operations

- Deployment allocation dashboard
- Allocation updates
- Strategy portfolio simulation
- Shared-capital simulation
- Active-set resolution
- Canonical deployment ownership
- Strategy route persistence

Matrix Replay

- Historical matrix replay
- Regime-aware route resolution
- Strategy lookup and execution
- Shared-capital portfolio outcome
- Equity curve
- Monthly and annual returns

- Replay attribution
- Persisted replay snapshots

Matrix Copilot

- Matrix review
- Matrix recommendations
- Scenario generation
- Alternative comparison
- Saved matrix proposals
- Proposal lookup and rename
- Proposal duplication
- Proposal approval and rejection
- Proposal restoration
- Matrix proposal decisions

9. Portfolio and Risk

Portfolio

- Paper portfolio view
- Portfolio construction view
- Positions and holdings
- Cash and portfolio value
- Paper trades
- Risk dashboard
- Portfolio recomputation
- Broker-to-ledger reconciliation
- Portfolio reconciliation records
- Ledger integrity endpoint

Portfolio Copilot

- Portfolio overview
- Natural-language portfolio questions
- Allocation recommendations
- Portfolio scenarios
- Alternative comparison
- Saved portfolio proposals
- Proposal approval and rejection
- Evidence and historical context assembly

Risk Controls

- Position sizing from account risk
- Stop-loss and take-profit calculation
- Daily loss limit
- Weekly loss limit
- Weekly drawdown limit
- Maximum position size
- Maximum total exposure
- Maximum sector exposure
- Maximum correlated exposure
- Correlation threshold
- Emergency kill switch
- Regime-based BUY blocking
- Automatic quantity reduction
- Trade risk classification

Risk Diagnostics

- Sector exposure summaries
- Return-correlation calculation
- Sector fallback for missing correlations
- Correlated-symbol groups
- Per-order violations
- Global safety violations
- Configured-limit reporting
- Position-aware risk explanation

10. Trading, Approvals and Ledger

Pre-Trade

- Pre-trade analysis endpoint
- Scan-opportunity lookup
- Market-regime context
- Position effect calculation
- Trade-risk calculation
- News and event summary
- Safety evaluation
- Recommendation generation
- Fail-closed error response
- Human-readable explanation

Approvals

- Approval request list and detail
- Approval creation and update
- Pending, approved, rejected, snoozed and executed states
- Approve, reject, snooze and resume actions
- Paper execution from approval
- Broker-paper execution from approval
- Execution requires an approved request
- Approval audit events

Paper Trading

- Internal paper orders
- Broker-paper orders
- Market and limit order types
- BUY and SELL sides
- Order preview
- Paper fills
- Paper positions
- Manual trade recording
- Trade list and creation
- Proposed-order list and update

Transaction Ledger

- Order created, approved, rejected and executed events
- BUY and SELL fill events
- Fee events
- Cash deposit and withdrawal events
- Position mark-to-market
- Position close events
- Portfolio reconstruction from ledger
- Cache rebuilding
- Ledger verification
- Broker account ledger sync

11. Broker Center

Broker Framework

- Broker adapter contract
- Broker adapter registry
- Provider selection
- Broker capability model
- Per-user broker configuration
- Encrypted broker secrets
- Secret-status reporting
- Connection logs
- Broker execution audits
- Broker order-event history

Moomoo

- Moomoo market-data provider
- Moomoo broker adapter
- Account discovery
- Account and position synchronization
- Simulated trading environment support
- Order preview and placement
- Order status synchronization
- Fill synchronization
- Order cancellation and modification
- Workspace execution lookup

IBKR

- IBKR host, port and client configuration
- Python-version compatibility check
- Socket reachability check
- Connection diagnostics
- Managed-account discovery
- Account-mode inference
- Paper-account enforcement
- Account summary, order and fill serialization
- IBKR status and test-connection interface

Broker Operations

- Broker health and preflight
- Account and capability views
- Order and fill lists
- Broker reconciliation
- Ledger synchronization
- Import broker order
- Mark order cancelled
- Order status refresh
- Manual order preview
- Execution-mode guard

Current Safety Boundary

- IBKR is documented as connection-diagnostic and does not place unrestricted live orders. Broker-paper paths and execution guards are implemented; live execution remains locked or supervised by configured modes.

12. Market Data

Market Data Hub

- Central provider hub
- Yahoo Finance provider
- Moomoo provider
- Scanner-result fallback provider
- Quote lookup
- Batch quotes
- Price history
- Chart data
- Provider health
- Market-data metrics

Data Handling

- Normalized market-data models
- Multiple chart ranges
- Intraday through monthly intervals
- TTL and stale-after policies
- Memory and Redis caches
- Provider fallback
- Persisted market-data snapshots
- Source and degraded-mode response headers
- Market metadata normalization

Market Data Interface

- Stock price charts
- Time-range selection
- Quote freshness display support
- Historical-context component
- Stock workspace
- Scanner and demo chart data
- Admin market-data metrics

13. Research Lab

Research Workspace

- Research project list, create, read and update
- Project archive and restore
- Research reports
- Project notes
- Research questions
- Investment theses
- Thesis review
- Project comparison
- Project change analysis
- Persisted research history

Research Copilot

- Market overview
- Natural-language questions
- Symbol research
- Sector research
- Scanner summary
- Watchlist summary
- Upcoming-event summary
- Deep research
- Market impact
- Portfolio impact
- Strategy impact
- Matrix impact
- Scanner impact
- Company comparison
- Theme research

Evidence

- Research context service
- Structured evidence records
- Source type and source identifier
- Source quality
- Symbol, sector and theme fields
- Metric name and value
- Timestamp and date range
- Interpretation and relevance score
- Historical-context display

14. AI Platform

AI Gateway

- Central AI service
- OpenAI provider integration
- Feature registry
- Prompt files and prompt cache
- Prompt versioning
- Prompt content hashing
- Feature-specific model configuration
- Structured output schemas
- Input validators
- Output validators
- AI invocation persistence

- AI diagnostics endpoint

Core AI Endpoints

- Stock analysis
- Portfolio analysis
- Scanner-result explanation
- Trade review
- News summary
- General chat
- Admin diagnostics

Integrated Copilots

- Strategy Copilot
- Portfolio Copilot
- Research Copilot
- Matrix Copilot
- Playbook recommendations
- Evidence-aware context
- Memory context integration

AI Controls

- Per-user AI rate limiting
- Internal AI rate limiting
- Feature governance
- Redaction service
- Input length and shape checks
- Schema-constrained responses
- Invocation audit data
- Metrics collection
- Cache TTLs for selected AI features
- Graceful provider fallback

15. Memory and Personalization

Memory Records

- Memory events
- Strategy, portfolio, matrix, research, trade, broker, approval, scanner, risk, AI and system categories
- Memory links
- Memory feedback
- Created-by classification
- Active, archived, excluded and delete-pending states
- Important-memory and timeline views

Embeddings

- Embedding configuration
- Embedding provider abstraction
- Vector repository
- Background embedding ingestion
- Pending, processing, completed, failed, stale and excluded states
- Retry and rebuild
- Mark stale
- Backfill script and admin endpoint
- Embedding diagnostics

Retrieval

- Vector similarity
- Text and entity signals
- Hybrid ranking
- Recency scoring
- Deduplication
- Context assembly
- Retrieval contract
- Retrieval audits
- Retrieval diagnostics

- Memory query endpoint

Personalization

- User memory profile
- Explicit preferences
- Behaviour patterns
- Personalization settings
- Preference and pattern actions
- Profile rebuild
- Personalization reset
- Memory export
- Delete account memory
- Remove personalized context
- Delete individual memory
- Useful, incorrect, corrected and outcome-based feedback

16. Alerts and Playbook

Alerts

- Scanner, approval, risk, portfolio, engine, broker, validation and system categories
- Info through critical severity
- Active, acknowledged, snoozed, resolved and expired states
- Alert rules
- Alert delivery records
- Alert health
- Mute and unmute category
- Acknowledge, snooze, resolve and dismiss actions

Telegram Delivery

- Telegram channel configuration
- Credential encryption
- Channel verification
- Test notification
- Delivery processing
- Pending, sending, sent, failed and retrying delivery states
- Local scanner alert persistence
- BUY-alert generation threshold

Playbooks

- Playbook dashboard
- Multiple playbooks
- Playbook creation
- Synchronization
- Snapshots
- Version creation
- Version comparison
- Version restore
- Version promotion
- Evidence rebuild
- Playbook export
- Export history

Playbook Intelligence

- Playbook runs
- Playbook insights
- AI recommendation generation
- Pending review
- Accept and reject recommendation
- Convert recommendation to strategy draft
- Recommendation evidence

- Regime detection
- Risk and safety
- Pre-trade analysis
- Paper execution
- Broker action bridges
- Scheduler and runtime artifacts

Data and Integrations

- PostgreSQL source of truth
- Optional Upstash Redis cache
- OpenAI integration
- Yahoo Finance
- Finnhub news
- Moomoo
- IBKR diagnostics
- Telegram delivery

Application Flow

- React cockpit
- Express API and security middleware
- Feature services
- Python quantitative engine
- Market-data and broker adapters
- PostgreSQL and cache
- Audit, memory, alerts and AI context

Current Platform Summary

Quant's Trade currently implements an integrated quantitative research and paper-trading cockpit. Its connected workflow spans market scanning, research, strategy design, backtesting, validation, deployment allocation, portfolio risk, approvals, broker-paper operations, outcome persistence, memory and playbook learning.

Verified Current Workflow

- | | |
|----|--|
| 01 | Scan markets and rank opportunities |
| 02 | Inspect technical, regime, news and position context |
| 03 | Research symbols, sectors and portfolio impact |
| 04 | Build, test and validate strategies |
| 05 | Route strategies through the deployment matrix |
| 06 | Evaluate portfolio and trade risk |
| 07 | Create and review an approval request |
| 08 | Execute internally or through a broker-paper path |
| 09 | Persist fills, ledger events and portfolio state |
| 10 | Feed outcomes into validation, memory and playbooks |